

Supplementary Material

Sustainable Hybrid Document-Routed Retrieval for Financial RAG: Resolving the Robustness–Precision Trade-off

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This document provides the per-group breakdowns and the full cross-company query subset referenced in the main text. All results were produced with the same code, indexes, models and query samples described in Sections 5–6 of the article; no new experiments are reported here.

1 Per-Group Breakdown of the Six-System Comparison

Section 6.4 of the article (Table 6) reports means aggregated over the full 1,500-query benchmark for all six systems. That benchmark consists of five disjoint groups of 300 queries each (G01–G05), sampled once from `Linq-AI-Research/FinDER` and held fixed across every system so that all comparisons use identical inputs. Table S1 gives the per-group values behind those means.

Metrics follow the definitions in Section 5.4 of the article. Answers are scored on a 1–10 scale by the GPT-4.1 evaluator against FinDER ground truth. **Avg Score** is the mean score over the group; $s=1$ is the failure rate (fraction of queries scored 1); $s\geq 8$ is the correctness rate; $s=10$ is the perfect-answer rate. The **Mean** row of each block is the value reported in Table 6 of the article.

The group-to-group ordering is stable: HDRR attains the highest average score and the lowest failure rate in all five groups, and CBR+LLM-Ctx (V-B) is second on average score in all five groups. Group G03 is the hardest or joint-hardest group for every one of the six systems, and G02/G04 the easiest, indicating that the residual inter-group variance reflects query difficulty rather than system-specific instability.

2 Cross-Company Query Subset

Section 6.5 of the article evaluates multi-document handling on a 25-query handcrafted subset, constructed entirely from companies already present in the indexed FinDER corpus. Table S2 lists the complete subset: 10 pairwise metric comparisons (Q01–Q10), 8 which-of- N ranking queries (Q11–Q18) and 7 risk-factor comparisons (Q19–Q25).

Table S2: The 25 handcrafted cross-company queries and their named tickers.

ID	Tickers	Query
Q01	AAPL,MSFT	Compare R&D expenditure between AAPL and MSFT for fiscal year 2023.

ID	Tickers	Query
Q02	NVDA,AMD	Compare gross margin between NVDA and AMD for fiscal year 2023.
Q03	WMT,TGT	Compare operating cash flow between WMT and TGT for fiscal year 2023.
Q04	KO,PEP	Compare free cash flow between KO and PEP for fiscal year 2023.
Q05	HD,LOW	Compare capital expenditure between HD and LOW for fiscal year 2023.
Q06	JPM,BAC	Compare net interest income between JPM and BAC for fiscal year 2023.
Q07	XOM,CVX	Compare capital expenditure between XOM and CVX for fiscal year 2023.
Q08	PFE,MRK	Compare R&D as a percentage of revenue between PFE and MRK for fiscal year 2023.
Q09	UNH,CI	Compare operating margin between UNH and CI for fiscal year 2023.
Q10	UPS,FDX	Compare operating margin between UPS and FDX for fiscal year 2023.
Q11	NVDA,AMD,INTC	Which of NVDA, AMD, INTC reported the highest R&D spend in fiscal year 2023?
Q12	WMT,COST,TGT	Which of WMT, COST, TGT reported the highest total revenue in fiscal year 2023?
Q13	DAL,UAL,AAL	Which of DAL, UAL, AAL reported the highest operating cash flow in fiscal year 2023?
Q14	RTX,LMT,NOC	Which of RTX, LMT, NOC reported the highest backlog at the end of fiscal year 2023?
Q15	JNJ,PFE,MRK	Which of JNJ, PFE, MRK reported the highest pharmaceutical-segment revenue in fiscal year 2023?
Q16	GS,MS,JPM	Which of GS, MS, JPM reported the highest investment-banking revenue in fiscal year 2023?
Q17	META,GOOG,NFLX	Which of META, GOOG, NFLX reported the highest advertising revenue in fiscal year 2023?
Q18	TSLA,GM,F	Which of TSLA, GM, F reported the highest automotive gross margin in fiscal year 2023?
Q19	AAPL,MSFT	How does AAPL's supply chain concentration risk compare to MSFT's in their fiscal year 2023 10-K?
Q20	JPM,GS	How does JPM's regulatory and capital-adequacy risk language compare to GS's in their fiscal year 2023 10-K?
Q21	AMZN,WMT	How does AMZN's labor and unionization risk disclosure compare to WMT's in their fiscal year 2023 10-K?
Q22	PFE,MRK	How does PFE's patent-expiration risk compare to MRK's in their fiscal year 2023 10-K?
Q23	NVDA,AMD	How does NVDA's China export-control risk disclosure compare to AMD's in their fiscal year 2023 10-K?
Q24	XOM,CVX	How does XOM's climate-transition risk compare to CVX's in their fiscal year 2023 10-K?
Q25	META,GOOG	How does META's antitrust and data-privacy regulatory risk language compare to GOOG's in their fiscal year 2023 10-K?

Table S1: Per-group results for all six systems on the five 300-query FinDER groups. Bold rows are the 1,500-query means reported in Table 6 of the article.

System	Group	n	Avg Score	$s=1$ (%)	$s \geq 8$ (%)	$s=10$ (%)
CBR	G01	300	5.85	24.3	47.7	12.0
	G02	300	6.17	21.7	51.7	16.0
	G03	300	5.64	25.7	44.0	14.0
	G04	300	6.21	21.0	51.3	14.7
	G05	300	6.21	20.0	50.7	12.3
	Mean	1,500	6.02	22.5	49.1	13.8
SFR	G01	300	6.53	9.3	44.0	7.7
	G02	300	6.61	9.3	46.3	8.7
	G03	300	6.11	12.0	36.7	10.7
	G04	300	6.67	9.7	50.7	10.0
	G05	300	6.32	11.0	42.3	5.3
	Mean	1,500	6.45	10.3	44.0	8.5
Agentic	G01	300	6.21	18.3	51.0	14.3
	G02	300	6.41	16.7	50.7	15.3
	G03	300	6.13	20.0	48.7	13.3
	G04	300	6.64	15.7	53.7	18.7
	G05	300	6.42	16.0	53.3	13.0
	Mean	1,500	6.36	17.3	51.5	14.9
CBR+Meta (V-A)	G01	300	6.73	12.7	54.3	17.0
	G02	300	7.18	10.0	61.3	18.7
	G03	300	6.59	12.7	54.0	15.3
	G04	300	7.21	9.7	61.7	20.0
	G05	300	6.87	13.3	58.3	16.0
	Mean	1,500	6.91	11.7	57.9	17.4
CBR+LLM-Ctx (V-B)	G01	300	7.20	10.7	62.3	19.7
	G02	300	7.33	9.7	68.0	19.0
	G03	300	7.10	10.3	61.0	22.3
	G04	300	7.37	7.7	65.0	22.0
	G05	300	7.32	8.3	66.3	18.3
	Mean	1,500	7.26	9.3	64.5	20.3
HDDR	G01	300	7.44	7.0	64.7	20.0
	G02	300	7.76	5.0	72.7	19.0
	G03	300	7.39	7.3	63.0	21.7
	G04	300	7.72	5.3	71.0	21.3
	G05	300	7.39	7.3	67.3	18.3
	Mean	1,500	7.54	6.4	67.7	20.1

Table S3 gives the per-query retrieval and answer coverage scores whose means appear in Table 7 of the article. Retrieval coverage is the fraction of tickers named in the query whose 10-K appears in the retrieved chunk set; answer coverage is the fraction that appear as standalone uppercase tokens in the generated answer. Both are objective and require no ground truth. HDRR’s routing coverage is 1.000 on all 25 queries and is therefore omitted from the table.

Table S3: Per-query retrieval (Ret.) and answer (Ans.) coverage on the cross-company subset.

ID	CBR		V-B		Agentic		HDRR	
	Ret.	Ans.	Ret.	Ans.	Ret.	Ans.	Ret.	Ans.
Q01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q02	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q03	1.00	1.00	0.50	1.00	1.00	1.00	1.00	1.00
Q04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q05	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q06	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q07	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q08	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q09	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q10	0.50	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q11	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q12	0.33	1.00	0.33	1.00	0.33	1.00	0.33	1.00
Q13	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q14	0.67	1.00	1.00	1.00	0.67	1.00	1.00	1.00
Q15	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q16	0.67	1.00	0.67	1.00	0.67	1.00	1.00	1.00
Q17	0.67	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q18	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q19	1.00	1.00	1.00	1.00	1.00	0.00	0.50	0.50
Q20	0.50	0.00	0.50	0.50	0.50	0.00	1.00	1.00
Q21	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q22	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q23	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q24	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Q25	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Mean	0.893	0.960	0.920	0.980	0.927	0.920	0.953	0.980

3 Data and Code Availability

The source code for all systems (indexing pipeline, retrieval components, routing layer and evaluation scripts), together with machine-readable versions of the tables above, is publicly available at <https://github.com/zhycheng614/financial-RAG-pipeline>.